

changing the behavior of the model, because the factor would cancel out when dividing by the mean fitness. Thus, as in the haploid model of natural selection, changes in allele frequencies among diploid organisms depend only on the relative success of each genotype under the assumptions we have made (e.g., no mutation, random mating, large population size).

Although the haploid and diploid models of natural selection are similar in form, the efficiency of natural selection differs substantially. As described in Problem 3.9, the diploid model requires twice the amount of selection to accomplish the same change in allele frequency as the haploid model. At an intuitive level, diploid selection is less effective because the benefits of the most fit allele are partially masked in heterozygotes by the expression of the least fit allele.

3.4 Models of Interactions among Species

Species do not exist in isolation from one another. The simple models of exponential and logistic growth fail to capture the fact that the growth of a species must depend on its interactions with other species. Do the species compete with or aid each other in obtaining resources? Do they fight for territories or provide a habitat for one another? Does a species consume or provide fodder for another species? A complete model describing every species interaction would be prohibitively complex except in the simplest of biological communities. Yet some insight can be obtained by focusing on specific types of interactions and how these interactions affect population growth. Below we focus on competition models (section 3.4.1) and consumer-resource models (section 3.4.2).

3.4.1 The Lotka-Volterra Model of Competition

Let us begin by introducing the Lotka-Volterra model of competition, which builds upon the one-species logistic model. In the logistic equation, intraspecific competition becomes more severe as the population size increases, resulting in a lower per capita growth rate (Figure 3.3). The Lotka-Volterra model extends the logistic model by allowing multiple species to compete for resources, resulting in both inter- and intraspecific competition.

To keep things simple, we consider only two competing species, whose numbers $n_1(t)$ and $n_2(t)$ change over time. The two species can have different intrinsic growth rates r_1 and r_2 , and different carrying capacities K_1 and K_2 . To account for competition between the two species, the Lotka-Volterra model assumes that each individual of species i experiences competition as if its own species had a population size of $n_i(t) + \alpha_{ij} n_j(t)$. (Here we use i to refer to either species 1 or 2, depending on which species is the current focus of our attention, and use j to refer to the competing species. This saves us from having to write down equations for both species each time.) The parameter α_{ij} is known as the *competition coefficient*, which converts the strength of competition exerted by an individual of species j on an individual of species i into the equivalent amount of competition that would be exerted if both individuals were of

Competition models describe how the number of individuals of each species changes when more than one species uses the same resource.

species i . Assuming that the reproductive factor of species i declines linearly with the amount of competition, we now have

$$R_i = \underbrace{(1 + r_i)}_{\text{intercept}} + \underbrace{\left(-\frac{r_i}{K_i}\right)}_{\text{slope}} (n_i(t) + \alpha_{ij} n_j(t))$$

Incorporating this reproductive factor into the discrete-time recursion equations $n_i(t + 1) = R_i n_i(t)$ and gathering together terms involving r_i , the Lotka-Volterra model becomes

$$n_1(t + 1) = n_1(t) + r_1 n_1(t) \left(1 - \frac{n_1(t) + \alpha_{12} n_2(t)}{K_1}\right), \quad (3.14a)$$

$$n_2(t + 1) = n_2(t) + r_2 n_2(t) \left(1 - \frac{n_2(t) + \alpha_{21} n_1(t)}{K_2}\right). \quad (3.14b)$$

Similarly, the continuous-time differential equations for the Lotka-Volterra model are

$$\frac{dn_1}{dt} = r_1 n_1(t) \left(1 - \frac{n_1(t) + \alpha_{12} n_2(t)}{K_1}\right), \quad (3.15a)$$

$$\frac{dn_2}{dt} = r_2 n_2(t) \left(1 - \frac{n_2(t) + \alpha_{21} n_1(t)}{K_2}\right). \quad (3.15b)$$

As a check, if the species do not interact in any way (i.e., $\alpha_{12} = \alpha_{21} = 0$), then equations (3.14) and (3.15) should reduce to the logistic equations (3.5a) and (3.5c), which indeed they do. Another check is provided by the fact that this model is also symmetric: which species is labeled 1 and which is labeled 2 is arbitrary. Thus, we can interchange the indices referring to species 1 and 2 in the recursion equation (3.14a) and regain equation (3.14b), and vice versa. The same holds for the differential equations (3.15a) and (3.15b).

Although we have been discussing competition, species j could facilitate the growth of species i , in which case α_{ij} would be negative. For example, species 1 could excavate holes that are used as nesting sites by species 2. The signs of α_{12} and α_{21} therefore reflect the relationship between the two species:

α_{12}	α_{21}	Relationship
–	–	Mutualistic
–	0	Commensal
0	–	Commensal
+	–	Parasitic
–	+	Parasitic
+	+	Competitive

Equations (3.14) and (3.15) can thus be used to describe the dynamics of populations experiencing many different types of species interactions.

3.4.2 Consumer-Resource Models

Consumer-resource models describe the dynamics of a resource and the dynamics of a population that consumes this resource.

Throughout the above, we have treated resources as constant. While this might be reasonable for some physical resources (e.g., light striking a patch of land or nutrients flowing down a river past a stationary aquatic community), in many cases the level of resources will be impacted by the growth of the species that we are modeling. For example, nutrients such as nitrogen might decline as a plant population grows, and the number of plankton might decline as a fish population grows. To account for such phenomena, we must construct models that explicitly track the dynamics of a resource as well as the population consuming the resource. Such models are referred to as *consumer-resource* models or as *predator-prey* models if the resource is itself an organism (a prey).

The general structure of a consumer-resource model is illustrated in Figure 3.8 for the case with one resource $n_1(t)$ and one consumer $n_2(t)$. From this flow diagram, we can derive the dynamical equations for this model in continuous time by summing the rate of change over all arrows for each circle (Recipe 2.4):

$$\begin{aligned}\frac{dn_1}{dt} &= f(n_1(t)) - g(n_1(t), n_2(t)), \\ \frac{dn_2}{dt} &= \varepsilon g(n_1(t), n_2(t)) - h(n_2(t)).\end{aligned}\tag{3.16}$$

Similar equations can be derived in discrete time, once we have specified the order in which events occur.

We have chosen to illustrate the flow diagram and derive these differential equations in very general terms by not specifying the functions $f(n_1)$, $g(n_1, n_2)$, and $h(n_2)$. The advantage of developing such a general model is that it makes it possible to derive many different consumer-resource models by appropriate choices for the functions $f(n_1)$, $g(n_1, n_2)$, and $h(n_2)$, as described next.

The term $f(n_1)$ represents the rate of change of the resource through means other than consumption (i.e., assuming $n_2(t) = 0$). This resource-renewal function

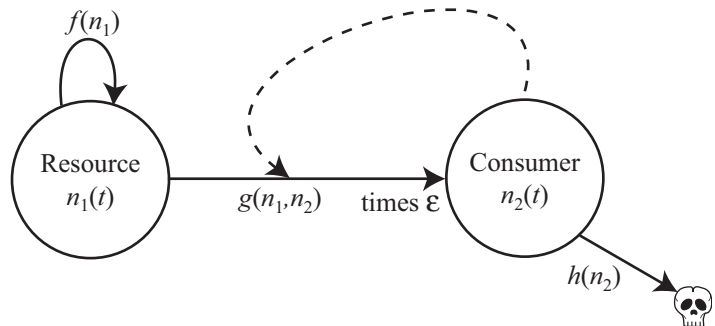


Figure 3.8: Flow diagram for consumer-resource model. A general flow diagram describing the levels of a resource n_1 and a consumer n_2 , where the rate of renewal of resources, $f(n_1)$, the rate of resource consumption, $g(n_1, n_2)$, and the rate of loss of consumers, $h(n_2)$, are left unspecified.

might involve inflow and outflow of abiotic resources (as in a chemostat; see Problem 2.4) or immigration, emigration, births, and deaths of biotic resources. Furthermore, the rate of change of the resource might depend on the current density of resources, or it might not. Table 3.3 lists some possible choices for $f(n_1)$. To simplify the notation, we drop the (t) notation, but we must remember that n_1 and n_2 are functions of time.

The term $g(n_1, n_2)$ represents the rate of consumption of the resource by the consumer. In the simplest case, a “mass-action” rate of consumption is assumed, as in the flu model of Chapter 2. That is, the total rate of contact between consumers and resources within the community is assumed to equal $c n_1 n_2$. At each contact, the probability that the consumer successfully uses the resource is a . Hence, $g(n_1, n_2) = a c n_1 n_2$, which is known as a “linear” or “type-I” consumption rate (sometimes referred to as a type-I functional response). Table 3.3 lists other common choices for $g(n_1, n_2)$. After having gathered a resource, the consumer converts it into the biomass needed for offspring production. In the flow diagram and in equation (3.16), the conversion factor by which resource units are turned into consumers is given by ϵ . For example, one prey might represent only a fraction ϵ of the resources needed to produce one predator.

Finally, $h(n_2)$ represents the rate at which the number of consumers changes in the absence of resources (i.e., assuming $n_1 = 0$). Typically, it is assumed that

TABLE 3.3

Consumer-resource models. Examples of functions that can be used in the consumer-resource model (3.16), where n_1 refers to the level of resources (e.g., number of prey) and n_2 refers to the level of consumers (e.g., number of predators).

Function	Description
$f(n_1) = \theta$	Inflow of resources at a constant rate
$f(n_1) = -\psi$	Outflow of resources at a constant rate
$f(n_1) = r n_1$	Constant per capita growth of resource species
$f(n_1) = r n_1 \left(1 - \frac{n_1}{K}\right)$	Per capita growth of resource species declines linearly with resource level (logistic)
$f(n_1) = r n_1 e^{-\alpha n_1}$	Per capita growth of resource species declines exponentially with resource level
$g(n_1, n_2) = a c n_1 n_2$	Linear (type I) rate of resource consumption
$g(n_1, n_2) = \frac{a c n_1}{b + n_1} n_2$	Saturating (type II) rate of resource consumption
$g(n_1, n_2) = \frac{a c n_1^k}{b + n_1^k} n_2$	Generalized (type III) rate of resource consumption
$h(n_2) = \delta n_2$	Constant per capita death rate of consumer
$h(n_2) = (\delta n_2) n_2$	Per capita death rate of consumer increases linearly with consumer population size

consumers die off at a constant per capita rate δ , so that $h(n_2) = \delta n_2$. But again, the rate of change of consumers might involve immigration, emigration, and even births if the consumer can utilize alternative resources. Furthermore, this rate might or might not depend on the current density of consumers. Table 3.3 lists two possible alternatives for $h(n_2)$.

By combining the functions $f(n_1)$, $g(n_1, n_2)$, and $h(n_2)$ in different ways, a large number of consumer-resource models are possible. Next, we present equations for two of these models, which we will return to in later chapters.

We start with the model that incorporates the simplest choices for each of these functions: a constant inflow (immigration) of resources at rate $f(n_1) = \theta$, a type-I rate of resource consumption $g(n_1, n_2) = a c n_1 n_2$, and a constant death rate of consumers $h(n_2) = \delta n_2$. Equations (3.16) then become

$$\begin{aligned}\frac{dn_1}{dt} &= \theta - a c n_1(t) n_2(t), \\ \frac{dn_2}{dt} &= \varepsilon a c n_1(t) n_2(t) - \delta n_2(t).\end{aligned}\tag{3.17}$$

Equation (3.17) can be used, for example, to model the inflow into a lake of a nutrient (e.g., nitrogen) that is required for and limits the growth of an algal species, where $n_1(t)$ and $n_2(t)$ represent the levels of nutrients and algae, respectively. By setting $\theta = 0$, equation (3.17) can be used to model the growth of a community when resources begin at some level $n_1(0)$ and are not replenished. Interestingly, as we shall see in Chapters 4 and 9, the logistic equation and equation (3.17) without immigration and death ($\theta = \delta = 0$) generate identical predictions about the level of consumers over time.

The second consumer-resource model that we consider is a classic one in ecology, known as the Lotka-Volterra predator-prey model. The only difference from the previous model (3.17) is that the resources are prey that undergo exponential growth in the absence of predators, $f(n_1) = r n_1$. Equations (3.16) then become

$$\begin{aligned}\frac{dn_1}{dt} &= r n_1(t) - a c n_1(t) n_2(t), \\ \frac{dn_2}{dt} &= \varepsilon a c n_1(t) n_2(t) - \delta n_2(t).\end{aligned}\tag{3.18}$$

As we shall see in Chapter 4, the Lotka-Volterra predator-prey model predicts that the number of prey and number of predators cycle over time, demonstrating that species interactions can lead to interesting dynamical behavior that might help explain the cyclic dynamics of several species (e.g., Fussmann et al. 2000; Krebs et al. 1995).

Although we have focused on continuous-time consumer-resource and predator-prey models, discrete-time models are straightforward to develop given the tools that you have now learned. As an example, we develop the discrete-time version of the Lotka-Volterra predator-prey model in Supplementary Material 3.2.